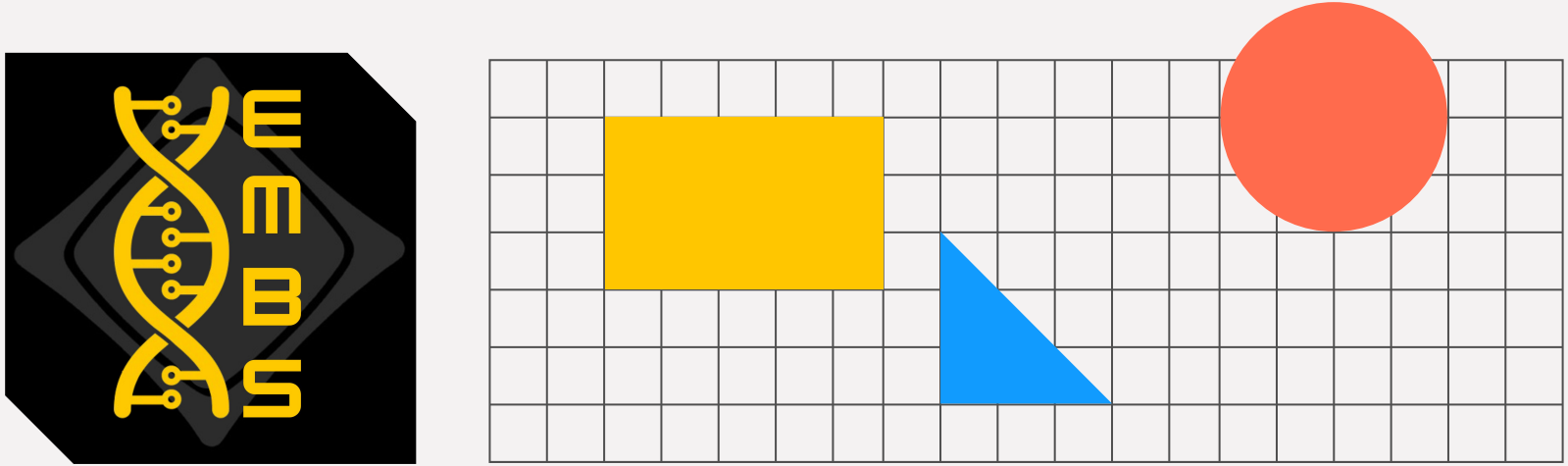
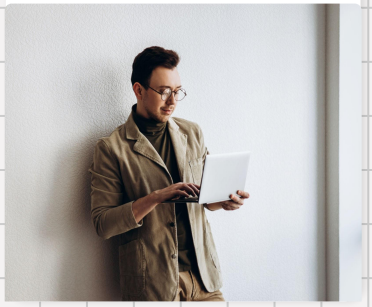
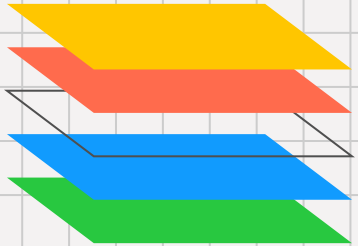
Brief Intro to Git and GitHub



About ~~Me~~ Git









Left: https://avatars.githubusercontent.com/u/18133?s=200&v=4
Right: https://commons.wikimedia.org/wiki/File:Octicons-mark-github.svg

Git is a version control tool commonly used in programming (software *and* hardware). This means scripts and other files can be edited and replaced without overwriting previous work. Having detailed documentation for ANY project is crucial, and Git is a powerful tool to ensure thorough documentation has been kept.

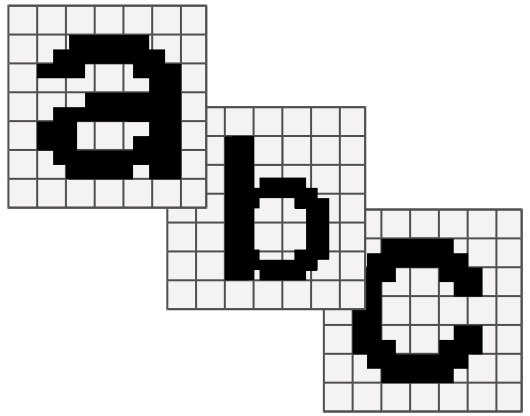
Git is the general name of this tool. GitHub, on the other hand, is a public website that stores Git-based projects.

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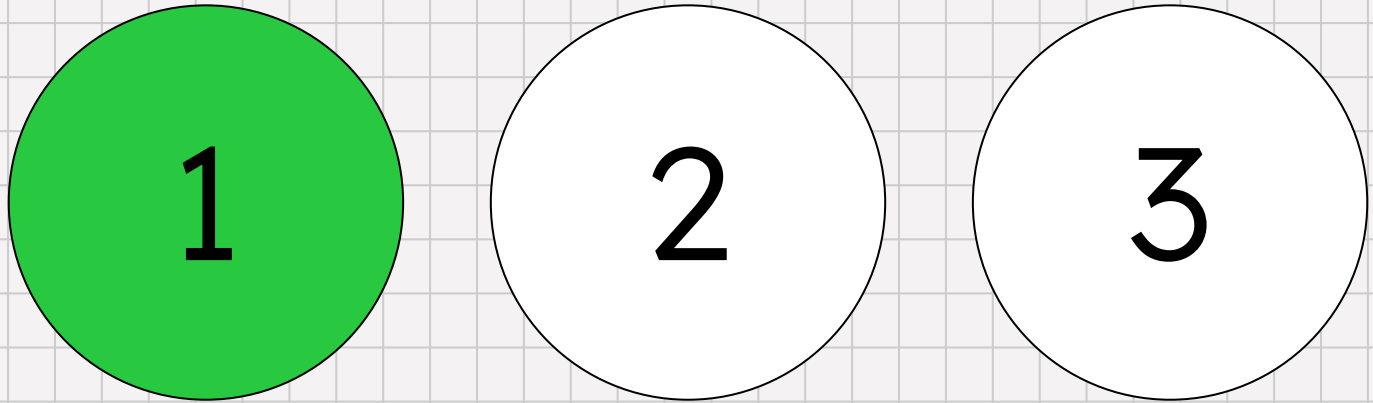
Learning Outcome

Why We're Here

Git can be intimidating to learn to someone unfamiliar with computer programming. This workshop will help guide you, step-by-step, and prepare you to work with one of the most powerful tools in programming.



Starting Out



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Setup

Download Instructions (should have completed before workshop):
https://phoenixnap.com/kb/how-to-install-git-windows

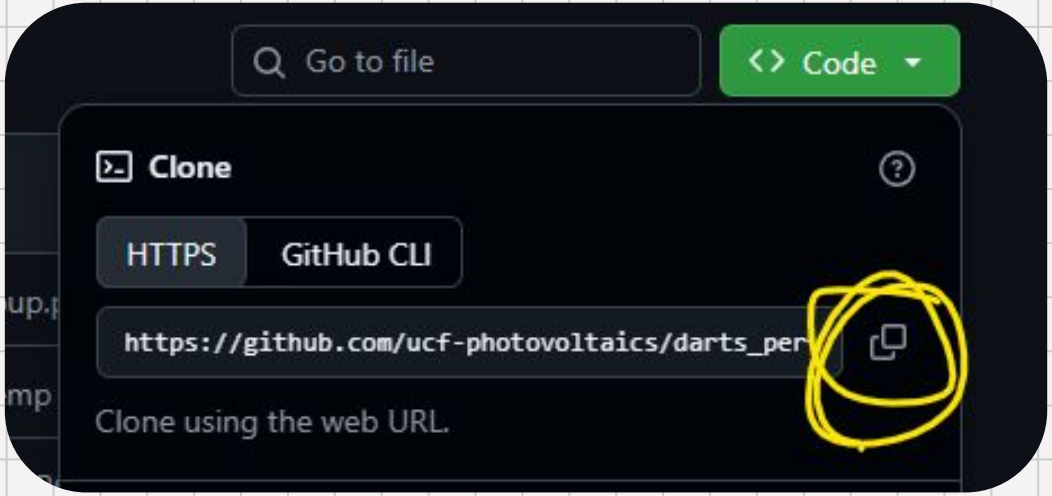
Open Git CMD, then initialize a local repository by typing in:

- cd “location-of-project-folder” (WITH quotes)
  - Ex. cd “C:\Users\froakie\Documents”
- mkdir name-of-new-folder (no quotes)
  - Ex. mkdir froakies-first-folder
- git init



https://win.gg/news/froakie-to-be-featured-in-the-pokemon-go-august-community-day/

Cloning a Repository



When you find your desired GitHub repo:

1. Select the green “Code” button and copy the URL
2. In Git CMD, type: git clone copied-GitHub-link (ex. git clone https://github.com/fuecoco/class_scheduler_project.git)

Pulling Files from a Remote Repo to Your Repo

To look at just 1 file instead of a whole repo:

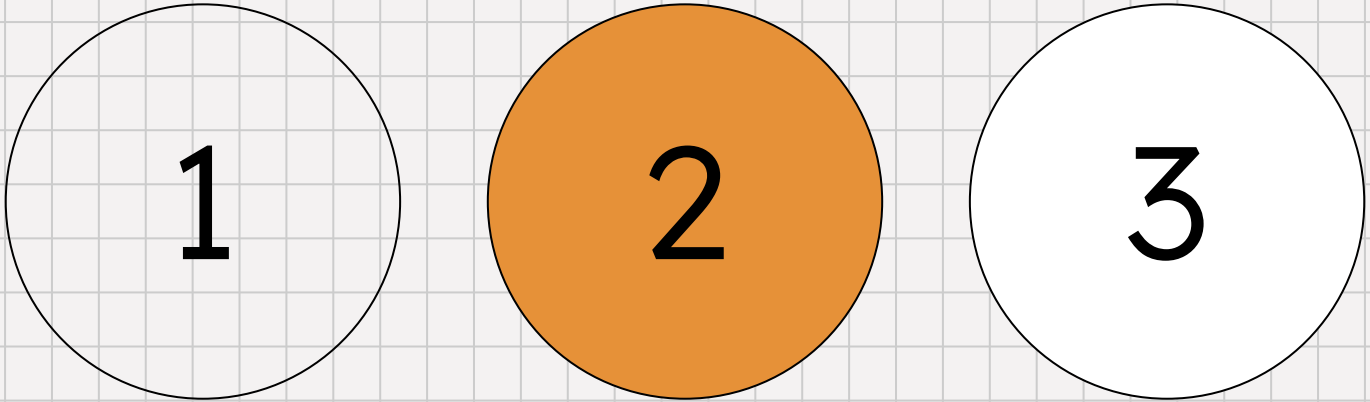
- git pull origin main
- git checkout -b branch-name
  - Makes a new branch

Other helpful commands:

- git branch (current branch)
- git switch branch-name (switch branches)

NEVER WORK ON THE MAIN BRANCH!!!

Editing Files



Commits

Checkpoints with different versions of code; saved in a commit history

General Psuedo-git(?) Sequence:

1. Check which files were modified
2. Add files to index (or staging area)
   a. Index – collects all changes; acts like a waiting area
3. Commit changes with a message
4. Verify that commit was successful



https://xkcd.com/1296/

Commit Commands

1. git status (standalone)
2. git add filename (individual files)
   - git add . (ALL files)
3. **git commit -m “[message]”**
   - If file is already tracked, can do git commit -a -m “[message]”
4. git log (shows all commits)

General Psuedo-git(?) Sequence:

1. Check which files were modified
2. Add files to index (or staging area)
   a. Index – collects all changes; acts like a waiting area
3. Commit changes with a message
4. Verify that commit was successful

<u>Example of a good commit message:</u>
git commit -m “Fixed blinking LED timing bug”



https://www.etsy.com/listing/193932343/froakie-plushie

Pushing Changes

Pushes to GitHub will back up/save your current code

To check for updates to the code:

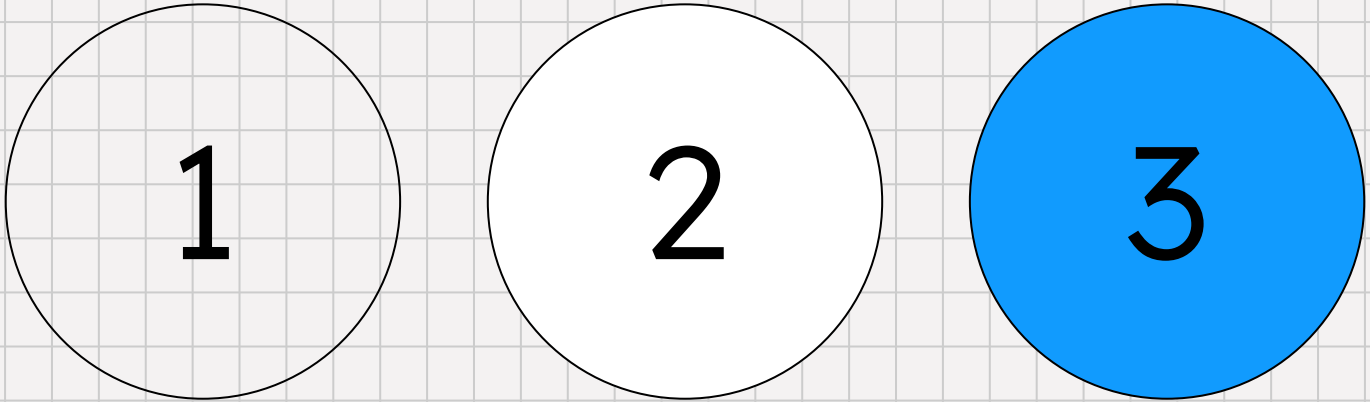
- git pull origin main

To update the remote repo:

- git push origin branch-name

Push changes DAILY in a workplace

Improving Files



Pull Requests and Issues

PRs

- Purpose: proposal to merge two branches in a repo
- Who Resolves: you (if you are granted approval to merge)
- Base = what to merge into, Compare = changed branch

Issues

- Purpose: point out new ideas or improvements for a repo
- Who Resolves: assignees (may or may not include you)
- Can add labels and/or milestones
- Close when finished

Formatting

Markdown (.md) text syntax allows for advanced formatting and emphasis of text for both PRs and issues (and technically commits)

text = *text*
text = **text**
text = ***text***

1. Ordered Item 1
2. Ordered Item 1

- Unordered A
- Unordered B

Heading 1 -> Heading 1

Heading 2 -> Heading 2

Heading 3 -> Heading 3 (etc.)

[link name](actual link)

![alt text](image.jpg)

Online resources have MANY guides on Markdown syntax beyond what is covered here!

So...what did we learn today?

Once you practice a bit, Git and GitHub are **not** as difficult as they seem!

- Git is a powerful tool for keeping project repositories **well-organized** for you and for others
- There are many commands that can generally be **grouped** by different tasks
- You work on your **own branch** when editing files, then you **push** your updated files to the main repo
- Pull requests and issues allow for teams to **communicate** where improvements can be made

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Tips

thank you

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Cloning a Repository

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Contact Me

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b
c